1 Saving an activation adds a rounding step



No probe

q → **normalize + rotate** (one kernel, full precision) → *query*



Probe attached

q → **normalize** → BF16 → **rotate** → *query′*

≠ last bits differ

2 One attention head reads the difference

Layer 35, head 6

`<|im_start|>` `user` Is a read-only probe … ? `assistant` `↵` → **Yes**

same offset in every prompt